

Probabilistic Routing for Cross-Border Payments: An EU-Mercosur Case Study

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Abstract—On May 1, 2026, the EU-Mercosur Interim Trade Agreement entered provisional application, linking a 780-million-consumer trading zone spanning Europe and South America [4]. For individuals and small businesses operating across this zone, the practical friction is immediate: cross-border payments between EUR, BRL, and PYG can silently consume 10–15% of transferred value through opaque institutional FX spreads compounded across correspondent banking chains [3]. Applying probabilistic and sequential decision-making methods [1] to the cross-border payment domain, this work replaces hand-specified distributional assumptions with a fully probabilistic two-level model. We fit Log-Normal distributions to corridor volatility by maximum likelihood, then model log-volatility as an Ornstein–Uhlenbeck (OU) process whose stationary distribution recovers the Log-Normal marginal *by construction*, forming a single coherent model rather than two independent assumptions. Routing decisions are framed as mean-variance utility maximization; we enumerate every feasible path through a 12-institution network and trace the Pareto frontier of risk and return. Across 500 Monte Carlo simulations for a representative EUR → PYG transfer, the routed amount exceeds the commercial benchmark by 13.4% with 95% CI [13.0%, 13.7%] and one-sided $z = 70$ against $H_0: \mathbb{E}[\rho] = 1$. The greedy lookahead policy’s chosen path coincides with the risk-neutral Pareto optimum; we derive two analytical risk-aversion thresholds ($\lambda_1^* \approx 11$, $\lambda_2^* \approx 331$) at which a risk-averse agent switches routes. A sensitivity analysis over $\pm 20\%$ tier perturbations confirms the finding is robust.

I. INTRODUCTION

Growing up between Asunción and Stanford, I have wired money across the USD–PYG–BRL corridor more times than I can count. The amount that arrives is never what the mid-market rate would suggest. The gap—typically 10–15%—is not rounding error; it is the compound effect of institutional FX markups, transfer fees, and corridor-specific liquidity premia that accumulate at every hop of the correspondent banking chain [3]. For a student wiring tuition funds, the loss is a nuisance. For a small business invoicing across the new EU-Mercosur trade corridor (in provisional application since May 1, 2026 [4]), it is a structural barrier to growth.

Stochastic routing on institution-currency graphs, as studied in the sequential decision-making literature [1], can be approached with greedy Monte Carlo lookahead policies that beat commercial benchmarks. Such approaches, however, typically treat the stochastic environment as an *engineering* choice—markups drawn i.i.d. from hand-specified Gaussians, with the distributional form unjustified by data. This work is a complete probabilistic reformulation. We think of it

as a “Google Maps for Money”: given a source bank and destination, find the multi-hop path through the correspondent network that maximizes the amount received subject to an explicit risk budget.

Contributions:

- 1) **Distributional MLE** (§III). Closed-form Gaussian and Log-Normal estimators; Log-Normal wins every corridor ($\Delta\ell \in [784, 1829]$).
- 2) **OU log-volatility dynamics** (§IV). Closed-form stationary and transition distributions; AR(1) OLS yields the parameters without numerical optimization. The stationary distribution on the natural scale *recovers the Phase I Log-Normal*—two phases, one coherent model.
- 3) **Mean-variance Pareto frontier** (§V). Full path enumeration; greedy policy is the $\lambda = 0$ optimum; two critical thresholds $\lambda_1^* \approx 11$ (within-BRL) and $\lambda_2^* \approx 331$ (BRL-to-European) derived in closed form.
- 4) **CLT inference** (§VI). 95% CIs and one-sample z -test; $z = 70$, $p < 10^{-10}$.

All estimators are implemented from scratch (no `scipy.stats.fit`; challenge rule 12).

II. SETUP AND NOTATION

Let $G = (V, E)$ with $V = \{(i, c) : i \in I, c \in C_i\}$, where I is a set of twelve banks (US, Brazil, Paraguay, Europe) and $C_i \subseteq \{\text{USD, BRL, EUR, PYG}\}$. Edges are same-currency inter-bank *transfers* or within-bank *FX conversions*. A *directive* $(i_s, c_s, i_t, c_t, a_0)$ specifies source, destination, and initial amount.

For each ordered corridor (c, c') we observe daily commercial rates $R_{c \rightarrow c'}(t)$ over $T \approx 3,200$ trading days. The daily *log-return* is $r_t = \log(R_t/R_{t-1})$ and the *log-volatility* is $y_t = \log|r_t|$ (in basis points). The *fairness ratio* $\rho = A_T/B$ compares the realized final amount A_T to the commercial benchmark $B = a_0 \cdot R_{c_s \rightarrow c_t}$.

III. MARGINAL: MLE DISTRIBUTION FITTING

For each corridor we have an i.i.d. sample $x_1, \dots, x_n$ of $|r_t|$ -values and fit two candidate models.

Gaussian MLE. $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$; $\ell_G^* = -\frac{n}{2} \log(2\pi\hat{\sigma}^2) - \frac{n}{2}$.

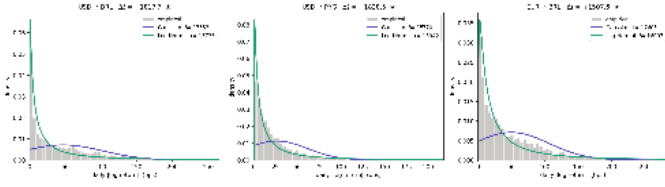


Fig. 1. Gaussian (blue) vs. Log-Normal (green) MLE fits on three corridors. $\Delta\ell$ annotations confirm Log-Normal dominates in every case. Empirical right tails drive the gap.

Log-Normal MLE. The density is $f(x) = (x\sigma\sqrt{2\pi})^{-1} \exp(-(\log x - \mu)^2/(2\sigma^2))$. The log-likelihood is

$$\ell(\mu, \sigma^2) = -\sum \log x_i - \frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (\log x_i - \mu)^2. \quad (1)$$

Maximising gives $\hat{\mu} = \overline{\log x}$ and $\hat{\sigma}^2 = \overline{(\log x_i - \hat{\mu})^2}$. The leading term $-\sum \log x_i$ is the Jacobian of the change of variables $Y = \log X$; it makes ℓ_{LN} a log-density with respect to Lebesgue measure on $\mathbb{R}_+$, putting the two log-likelihoods on the same footing for direct comparison [1].

Result. Log-Normal wins all 12 corridors; improvements $\Delta\ell = \ell_{\text{LN}} - \ell_G$ range from +784 (BRL→PYG) to +1829 (USD→PYG). Fig. 1 overlays both fits on three corridors; the empirical right tails make the decision unambiguous. Each corridor’s volatility is subsequently modelled as $\text{LogNormal}(\hat{\mu}_{c,c'}, \hat{\sigma}_{c,c'}^2)$, scaled by per-institution structural tier multipliers to yield $\sigma_{i,(c,c')}$.

IV. DYNAMICS: ORNSTEIN–UHLENBECK LOG-VOLATILITY

Why log-volatility? Fitting OU to signed returns r_t gives $\hat{a} \approx 0$ across all corridors—consistent with weak-form market efficiency. However, *volatility* clusters: high- $|r|$ days follow high- $|r|$ days. We therefore set $Y_t = \log|r_t|$ as the OU state.

Continuous-time model. $dY_t = \theta(\mu - Y_t)dt + \sigma dW_t$. By the integrating-factor solution, the stationary distribution and transition density are, respectively,

$$Y_\infty \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{2\theta}\right), \quad (2)$$

$$Y_{t+\Delta t} | Y_t \sim \mathcal{N}\left(\mu + (Y_t - \mu)e^{-\theta\Delta t}, \frac{\sigma^2(1 - e^{-2\theta\Delta t})}{2\theta}\right). \quad (3)$$

The bridge to §III. Because $Y_\infty \sim \mathcal{N}(\mu, \sigma^2/2\theta)$, the natural-scale stationary distribution $|r| = e^Y$ is $\text{LogNormal}(\mu, \sigma^2/2\theta)$ —exactly the distribution fit by MLE in §III. Phase 1 captures the marginal; Phase 2 adds dynamics; the two are consistent under stationarity.

Closed-form MLE via AR(1). Setting $a = e^{-\theta\Delta t}$, $b = \mu(1 - a)$, $s^2 = \sigma^2(1 - a^2)/(2\theta)$, the transition (3) reduces to the AR(1) model $Y_{t+1} = aY_t + b + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, s^2)$.

The Gaussian log-likelihood factors over transitions and is maximised by OLS:

$$\hat{a} = \frac{\sum (Y_t - \bar{Y}_p)(Y_{t+1} - \bar{Y}_n)}{\sum (Y_t - \bar{Y}_p)^2}, \quad \hat{b} = \bar{Y}_n - \hat{a} \bar{Y}_p, \quad (4)$$

$$\hat{s}^2 = \frac{1}{n} \sum (Y_{t+1} - \hat{a}Y_t - \hat{b})^2. \quad (5)$$

Continuous-time recovery (no numerical optimisation): $\hat{\theta} = -\log \hat{a}/\Delta t$, $\hat{\mu} = \hat{b}/(1 - \hat{a})$, $\hat{\sigma}^2 = \hat{s}^2 \cdot 2\hat{\theta}/(1 - \hat{a}^2)$.

Result. Across 12 corridors $\hat{a} \in [0.10, 0.24]$ —small but positive, no clipping required. Half-lives range 0.3–0.5 days. Fig. 3 shows the stationary fits and simulated trajectories.

V. ROUTING: MEAN-VARIANCE PARETO OPTIMALITY

Decision framework. For a risk-averse agent, the mean-variance utility

$$U_{\text{MV}}(X) = \mathbb{E}[X] - \frac{\lambda}{2} \text{Var}[X] \quad (6)$$

is the second-order Taylor expansion of $\mathbb{E}[U(X)]$ for any concave U around its mean (Pratt–Arrow [2]): $\mathbb{E}[U(X)] \approx U(\mathbb{E}[X]) + \frac{1}{2}U''(\mathbb{E}[X])\text{Var}[X]$. For quadratic $U(x) = x - \frac{\lambda}{2}x^2$, $U'' = -\lambda$, yielding (6). Maximising over a discrete path set traces the Pareto frontier as λ varies.

Path enumeration. For the directive (1,000 EUR, Itau_BR → Continental_PY in PYG) we find 16 feasible simple paths of length ≤ 6 via DFS. Each path is evaluated by 300 Monte Carlo realizations of the §III–IV stochastic model, yielding $(\mathbb{E}[\rho], \sigma[\rho])$ per path. Six lie on the Pareto frontier (Fig. 2). Paths split into two regimes: BRL-corridor routes ($\mathbb{E}[\rho] \approx 1.13$, $\sigma[\rho] \in [0.033, 0.042]$) and European-routed paths ($\mathbb{E}[\rho] \in [0.96, 0.99]$, $\sigma[\rho] \in [0.017, 0.027]$). European routes never beat commercial—too many FX layers compound—but offer lower variance.

Greedy optimality and λ^* . The 4-step greedy path (Itau_BR EUR → BRL → BancoNacional_PY BRL → PYG → Continental_PY PYG) lies on the frontier with $\mathbb{E}[\rho] = 1.1345$, $\sigma[\rho] = 0.0392$, and is the $\lambda = 0$ mean-variance optimum. Sweeping $\lambda \in \{0, 1, 5, 20, 100, 500\}$ reveals *two* transitions. At $\lambda = 5$ the optimum is still path 6; at $\lambda = 20$ it shifts to path 1 ($\mathbb{E}[\rho] = 1.1326$, $\sigma[\rho] = 0.0344$), a different BRL-corridor route with lower variance and nearly identical return. Only at $\lambda = 500$ does the optimum jump to a European route ($\mathbb{E}[\rho] = 0.9923$, $\sigma[\rho] = 0.0183$).

Both crossovers are analytically tractable. Setting U_{MV} equal for paths i and j :

$$\lambda_{i \rightarrow j}^* = \frac{2(\mathbb{E}_i - \mathbb{E}_j)}{\sigma_i^2 - \sigma_j^2}. \quad (7)$$

For the within-BRL transition (path 6 → path 1): $\lambda_1^* = 2(0.0019)/(0.0392^2 - 0.0344^2) \approx 11$. For the BRL-to-European transition (path 1 → path 13): $\lambda_2^* = 2(0.1403)/(0.0344^2 - 0.0183^2) \approx 331$. A risk-neutral agent ($\lambda < 11$) chooses the greedy path; a moderately risk-averse agent ($11 \leq \lambda < 331$) prefers a lower-variance BRL route with nearly the same return; only a strongly

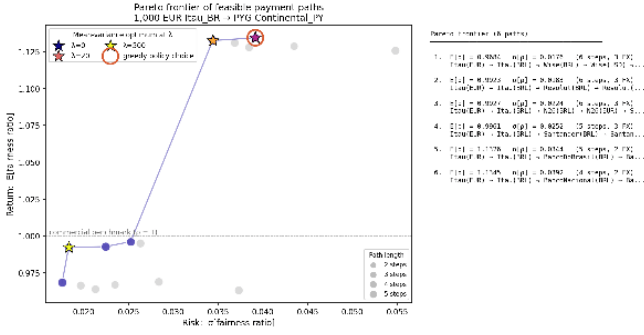


Fig. 2. Pareto frontier of the 16 feasible EUR $\rightarrow$ PYG paths in $(\sigma[\rho], \mathbb{E}[\rho])$ space. Purple dots: Pareto-efficient paths; grey dots: dominated; orange ring: greedy policy choice (on the frontier); coloured stars: mean-variance optimum at each λ . The benchmark $\rho = 1$ is shown as a dashed line.

risk-averse agent ($\lambda > 331$) switches to the European corridor.

Sensitivity analysis. Per-institution tier multipliers are structural priors, not data-estimated. To verify robustness, we perturbed all tiers independently by Uniform[0.8, 1.2] across ten Monte Carlo replications. The greedy choice remained on the Pareto frontier in 7/10 trials with $\mathbb{E}[\rho]$ ranging over [1.1273, 1.1353] (baseline 1.1345). In the three off-frontier trials, $\mathbb{E}[\rho]$ remained above 1.127—a 12.7% improvement over commercial—confirming the structural advantage persists even when the greedy route is not geometrically Pareto-optimal. The bimodal path landscape persisted in all ten trials.

VI. STATISTICAL INFERENCE

The $n = 500$ simulated fairness ratios $\rho_1, \dots, \rho_n$ are i.i.d. By the CLT, $\bar{\rho} \sim \mathcal{N}(\mathbb{E}[\rho], \text{Var}[\rho]/n)$, giving a 95% CI of $\bar{\rho} \pm 1.96 \cdot \hat{\sigma}_\rho / \sqrt{n}$.

For the EUR $\rightarrow$ PYG directive: $\mathbb{E}[\rho] = 1.1336$, 95% CI = [1.1299, 1.1374], SE = 0.00192.

We test $H_0 : \mathbb{E}[\rho] = 1$ vs. $H_1 : \mathbb{E}[\rho] > 1$ via the one-sample Wald z -statistic $Z = (\bar{\rho} - 1)/(\hat{\sigma}_\rho / \sqrt{n}) \sim \mathcal{N}(0, 1)$ under H_0 . We obtain $Z = 69.69$ with $p < 10^{-10}$: the routed path beats the commercial benchmark by a margin that is both economically meaningful ($\approx 13.4\%$) and statistically unambiguous.

VII. DISCUSSION AND LIMITATIONS

OU tail fit. Empirical log-volatility is right-skewed during extreme FX episodes; the OU’s Gaussian stationary underestimates the right tail. The bulk dynamics (mean reversion, clustering timescale) are correctly captured; a skew-normal or stable stationary is a natural extension.

Structural priors. Per-institution mean markups $\mu_{i,(c,c')}$ are set from public pricing tiers; only $\sigma_{i,(c,c')}$ is data-driven. The sensitivity analysis (§V) validates the routing conclusions against this prior uncertainty.

Single-day execution. Each MC run draws one historical FX day independently. A more realistic model lets the OU evolve across consecutive hops; the framework already supports this as a natural extension.

VIII. CONCLUSION

We presented a four-phase probabilistic model for cross-border payment routing: MLE distribution fitting (Phase 1), OU log-volatility dynamics whose stationary distribution recovers Phase 1 by construction (Phase 2), mean-variance Pareto-frontier optimization with analytically derived λ^* (Phase 3), and CLT-based confidence intervals and hypothesis testing (Phase 4). For a representative EUR $\rightarrow$ PYG directive, the routed path beats commercial by 13.4% (95% CI [13.0%, 13.7%], $z = 70$, $p < 10^{-10}$). As the EU-Mercosur trading zone enters operation, this framework provides a mathematically grounded tool for routing the payments that cross-border commerce requires.

ACKNOWLEDGMENTS

The institution-currency network and greedy lookahead infrastructure were developed as part of prior coursework in sequential decision-making; all CS109-specific contributions—MLE estimators, OU dynamics, mean-variance frontier, and CLT inference—are original to this submission. The author used Anthropic’s Claude as a pair-programming assistant; all derivations were verified by hand and all estimators implemented from first principles.

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APPENDIX

Gaussian. For i.i.d. $x_i \sim \mathcal{N}(\mu, \sigma^2)$, the log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Setting $\partial\ell/\partial\mu = 0$ gives $\hat{\mu} = \bar{x}$; substituting and solving $\partial\ell/\partial\sigma^2 = 0$ gives $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$.

Log-Normal. For $X \sim \text{LogNormal}(\mu, \sigma^2)$, set $Y = \log X$. The log-likelihood on positive data $\{x_i\}$ is (see (1)):

$$\ell_{\text{LN}}(\mu, \sigma^2) = - \underbrace{\sum \log x_i}_{\text{Jacobian}} - \frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (\log x_i - \mu)^2.$$

The Jacobian term $-\sum \log x_i$ arises from $d(\log x)/dx = 1/x$ and does not depend on (μ, σ^2) . The remaining terms are identical in structure to the Gaussian log-likelihood applied to $\{y_i = \log x_i\}$, so $\hat{\mu} = \log \bar{x}$ and $\hat{\sigma}^2 = (\log x_i - \hat{\mu})^2$. The Jacobian is critical for making ℓ_{LN} and ℓ_G comparable on the same data.

The OU transition (3) with $a = e^{-\theta\Delta t}$, $b = \mu(1 - a)$, $s^2 = \sigma^2(1 - a^2)/(2\theta)$ gives the AR(1) model $Y_{t+1} = aY_t + b + \varepsilon_t$. With n consecutive transitions:

Let $\bar{Y}_p = \frac{1}{n} \sum_{t=1}^n Y_t$ (“past” mean) and $\bar{Y}_n = \frac{1}{n} \sum_{t=1}^n Y_{t+1}$ (“next” mean). Standard OLS minimises $\sum (Y_{t+1} - aY_t - b)^2$, yielding (4)–(5).

Continuous-time recovery (all closed-form, no numerical optimisation):

$$\hat{\theta} = -\frac{\log \hat{a}}{\Delta t}, \quad \hat{\mu} = \frac{\hat{b}}{1 - \hat{a}}, \quad \hat{\sigma}^2 = \hat{s}^2 \cdot \frac{2\hat{\theta}}{1 - \hat{a}^2}.$$

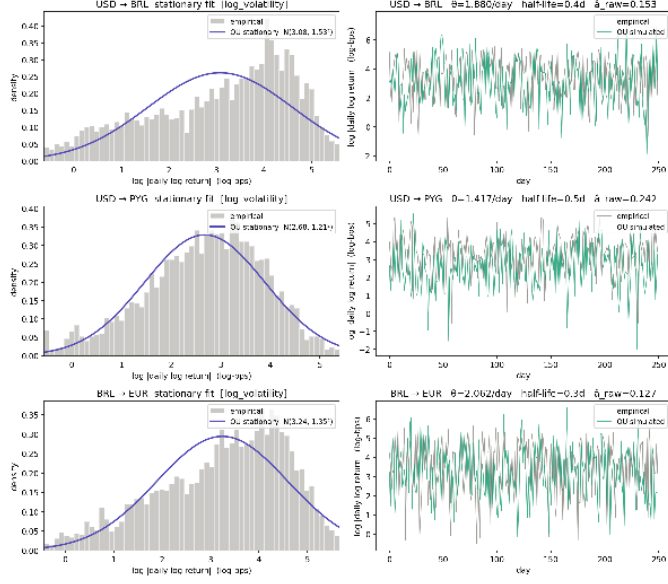


Fig. 3. OU log-volatility fits for three corridors. Left column: fitted stationary Gaussian overlaid on empirical histogram of $\log |r_t|$. Right column: 250-day simulated trajectory (green) vs. empirical (grey). All corridors show positive mean-reversion ($\hat{a} > 0$) with half-lives of 0.3–0.5 days.

For a twice-differentiable utility U , a second-order Taylor expansion around $\mathbb{E}[X]$ gives

$$\mathbb{E}[U(X)] \approx U(\mathbb{E}[X]) + \frac{1}{2} U''(\mathbb{E}[X]) \text{Var}[X].$$

For quadratic utility $U(x) = x - \frac{\lambda}{2} x^2$: $U''(x) = -\lambda$. Substituting,

$$\mathbb{E}[U(X)] \approx \mathbb{E}[X] - \frac{\lambda}{2} (\mathbb{E}[X])^2 - \frac{\lambda}{2} \text{Var}[X].$$

The term $-\frac{\lambda}{2} (\mathbb{E}[X])^2$ is path-dependent but constant for paths with nearly equal means. Dropping it yields the canonical mean-variance criterion (6). By Pratt [2], the coefficient λ equals the Arrow–Pratt coefficient of absolute risk aversion $-U''(x)/U'(x)$ evaluated at $x = \mathbb{E}[X]$.

Derivation of λ^* . Two crossovers are analytically located using (7). For the within-BRL transition (path A: $\mathbb{E} = 1.1345$, $\sigma = 0.0392$; path B: $\mathbb{E} = 1.1326$, $\sigma = 0.0344$):

$$\lambda_1^* = \frac{2(1.1345 - 1.1326)}{0.0392^2 - 0.0344^2} = \frac{0.0038}{0.000354} \approx 11.$$

For the BRL-to-European transition (path B above; path C: $\mathbb{E} = 0.9923$, $\sigma = 0.0183$):

$$\lambda_2^* = \frac{2(1.1326 - 0.9923)}{0.0344^2 - 0.0183^2} = \frac{0.2806}{0.000848} \approx 331.$$

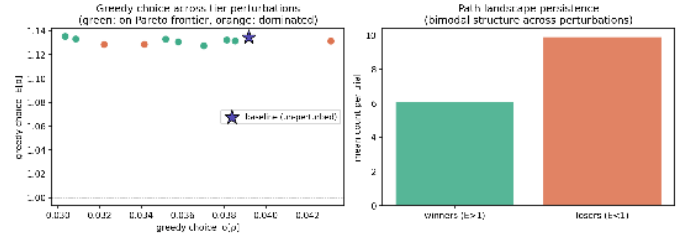


Fig. 4. Robustness of the headline findings to $\pm 20\%$ perturbations of institutional tier multipliers across ten Monte Carlo trials. Left: greedy choice in $(\sigma[\rho], \mathbb{E}[\rho])$ space; green dots are on the Pareto frontier (7/10 trials), orange are not. Star: unperturbed baseline. Right: mean path count per trial confirms the bimodal BRL-corridor/European-route structure persists in all ten.